**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 142 · Lesson 142 · Networks: LANs, WANs, Standards and Protocols

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Which ACID property keeps committed transaction changes after a crash?

Q2. How many bits are in one byte?

Q3. What data structure uses LIFO?

Q4. What is a trace table used for?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

LAN — Network spanning a relatively small geographical area.

WAN — Network connecting networks over larger distances.

protocol — Agreed rules for communicating between systems.

standard — Shared specification that enables interoperability.

## New learning

Identify scope — LAN connects local devices; WAN connects distant sites.

Interconnect devices — Switches connect local devices; routers forward traffic between networks.

Use open rules — Protocols and standards let different equipment communicate.

## Worked example

Two classrooms share printers in the same school. Is this LAN or WAN, and why are protocols necessary?

1. Same school buildings — Identify the relevant information and make a prediction.

2. Local switched/Wi-Fi network — Explain the change using correct technical terms.

3. Standard protocols — A school LAN connects local devices. Ethernet/Wi-Fi standards support compatible links and IP-based protocols allow data exchange. A router becomes relevant for communication beyond the local IP network.

# We Do · Practise with guidance

A trust connects schools 40 miles apart using provider links. Classify the network.

Separate school LANs

Identify the relevant information and make a prediction.

Router/provider link

Explain the change using correct technical terms.

Connected WAN

It is a WAN connecting multiple site LANs. Routers and service-provider infrastructure carry traffic between sites; secure connections and availability matter.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Differentiate LAN and WAN with examples. [6]

### Question 2

Explain why protocols and standards matter when devices use different manufacturers. [6]

### Question 3

A school requires a shared printer, wireless access and broadband internet. Explain suitable network devices and their functions. [8]

## Exit Ticket · Answer before checking

A WAN is defined by having more devices than a LAN.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Durability.

Q2. Eight.

Q3. Stack.

Q4. Tracking variable state changes as an algorithm runs.

### You Do pages · Independent questions

Q1. LAN limited local scope [2] and example [1], WAN wider link of networks [2] and example [1].

Q2. Common data formats/behaviours [2], compatibility/interoperability [2], reliable predictable exchange [2].

Q3. Switch [2], wireless access point [2], router [2], justification of links/traffic [2].

### Exit Ticket · Model answer

False. The distinction is geographical scope/connection of distant networks.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**